

Program : Diploma in Artificial Intelligence & Machine Language	
Course Code: 6349A	Course Title: Deep learning lab
Semester: 6	Credits: 1.5
Course Category: Program elective course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

Course Prerequisites:

Topic	Course Title	Semester
Basic programming skills	Introduction to IT Systems Lab	1
	Problem Solving & Programming	2
	Python Programming Lab	3

Course Outcomes

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize TensorFlow	11	Applying
CO2	Implement Multilayer Perceptrons (MLPs)	9	Applying
CO3	Implement Convolutional neural networks	12	Applying
CO4	Implement Deep learning application for Reinforcement learning	10	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize TensorFlow		
M1.01	Set up the development environment with TensorFlow.	2	Applying
M1.02	Familiarize basic TensorFlow operations: tensors, operations, sessions	3	Applying
M1.03	Build and execute a simple computational graph	3	Applying
M1.04	Building and train a simple neural network	3	Applying
CO2	Implement Multilayer Perceptrons (MLPs)		
M2.01	Implement a MLP for Binary Classification.	4	Applying
M2.02	Implement a MLP for Multi-Class Classification.	5	Applying
	Lab Exam – I	1.5	
CO3	Implement Convolutional neural networks		
M3.01	Write a program for character recognition using CNN	3	Applying

M3.02	Write a program to predict a caption for a sample image using CNN	3	Applying
M3.03	Implement Image Classification with CNN.	3	Applying
M3.04	Implement Text Classification with Word Embeddings.	3	Applying
CO4	Deep learning application for Reinforcement learning and NLP		
M4.01	Implement Text Generation with RNN.	4	Applying
M4.02	Open ended experiments	6	Applying
	Lab Exam – II	1.5	

**** Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

Text / Reference:

T/R	Book Title / Author
R1	Advanced Deep Learning with TensorFlow 2 and Keras Apply DL, GANs, VAEs, deep RL, unsupervised learning, object detection and segmentation, and more, Second Edition Rowel Atienza
R2	Deep Learning with Python. Second edition Francis Chollet
R3	"Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron.

Online Resources:

Sl. No.	Website Link
1	https://onlinecourses.nptel.ac.in/noc20_cs95/preview